MIL-C-50902A (PA)
9 August 1974
SUPERSEDING
MIL-C-50902
14 January 1972

MILITARY SPECIFICATION

CARTRIDGE, 152MM, TP-T, M411A3 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This Military Specification covers the loading, assembling and packing for one type of target practice cartridge designated as Cartridge, 152MM, TP-T, M411A3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-P-223 - Powder, Black
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of
MIL-I-50867 - Ignition Element, Electric; Loading, Assembling and Packaging
MIL-I-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-60312 - Parts, Molded, Plastic Foam, Polystyrene (For Use With Ammunition)

STANDARDS

MILITARY

MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-453 - Inspection, Radiographic

DRAWINGS

PICATINNY ARSENAL

FSC: 1320

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PRODUCT AND PACKAGE DRAWINGS

9255465 - Packing and Marking for Cartridge, 152MM, Series M409, Series M411 and Series Cannister M625.
9212218 - Igniter Assembly
9260121 - Cartridge Assembly with Bag, Barrier, Elastomeric
9249028 - Ignition Element, Electric
9266944 - Cartridge, 152MM, TP-T, M411A3

INSPECTION EQUIPMENT DRAWINGS

9201156 - Gage, Profile and Alignment
9201157 - Gage, Alignment Check
9201752 - Gage, Limit Check
9201766 - Fixture, Shell Extracting

3. REQUIREMENTS

3.1 General.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Cartridge.-The cartridge shall comply with all requirements specified on Drawing (dwg.) 9266944 and with all requirements specified in applicable specifcations.

3.3 Torque Requirements

3.3.1 Assembly Torque.-Thread joints shall withstand subsequent reapplication of the minimum (min.) assembly torque specified on the applicable drawing, without moving, while thread sealing compounds (if applicable) are wet.

3.3.2 Disassembly Torque.-Thread joints shall withstand application of min. disassembly torque specified on the applicable drawing, without moving, while thread sealing compounds (if required) are wet.

3.4 Moisture Content of Black Powder in Igniter Assemblies.-The moisture content of the black powder loaded into igniter bags at time of loading shall not exceed 0.70 percent (%).

3.5 Proving Ground.-The cartridge shall comply with the following requirements.

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3.5.1 High Temperature Test.-The maximum (max.) chamber pressure for any round fired at +125°F shall not exceed 115% of the upper pressure limit specified for the propellant (44,200 psi Piezo Equivalent Pressure).

3.5.2 Safety.-There shall be no audible hangfires, or indication of metal parts separation or breakup in the gun bore or in flight prior to impact during any of the proving ground tests conducted.

3.5.3 Residue.-The cartridge case components shall function satisfactorily without excessive or smoldering residue.

3.5.4 Muzzle Velocity.-The corrected muzzle velocity when fired at 70 degrees Fahrenheit (°F) shall not be greater than 2270 feet per second (fps) or less than 2210 fps, and the standard deviation of the lot shall not be greater than 10.1 fps.

3.5.5 Tracer Functioning.-The tracer shall ignite and there shall be visible evidence of burning to 1000 meters.

3.6 Workmanship

3.6.1 All parts shall be free of chips, dirt, grease, rust, burrs, cracks and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.6.2 The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such paintings have been marred or obliterated. (Bare spots in painting, other than marking, to total not more than 1/4 square inch in aggregate, are allowable.

3.7 Ignition Element Circuit Continuity.-The resistance of the ignition element after insertion into the cartridge case base shall not be greater than 1.0 ohm.

3.8 Drying Time.-After assembly of cartridge case to cartridge case body, the cartridge shall be cured in the conditioning bay in accordance with the drying time specified on the applicable drawing.

3.9 Moisture Content of Cartridge Case.-At the time of assembly of elastomeric barrier bag to the cartridge, the average moisture content of the case body and case base shall not exceed 1.5% and no individual reading shall exceed 2.0%.

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3.10 Pre-use Inspection of Cartridge Case Components.-The gap between the cartridge case body outside diameter and the cartridge case base inside diameter shall not exceed .010 inch for a depth of .0625 inch (max.).

3.11 Moisture Content of Cushion-half.-At the time of assembly with the cartridge, the moisture contnent of the cushion-half shall not exceed 0.4%.

3.12 Cavitation.- (Dwg. 9252343 applies).

3.12.1 Maximum (max.) projected areas and dimensions of cavities in each of the segments (See Figure 1) shall be in accordance with Table I.

TABLE I

SEGMENT A

Sum of areas of all cavities (square inches)	Max. dimension of any cavity (inches)
2.0	2.0

REMAINDER OF PROJECTILE

Sum of areas of all cavities (square inches)	Max. dimension of any cavity (inches)
1.5	1.0

3.12.2 Cavities Extending into Two Segments.-Cavities extending into two segments shall meet the requirements of both segments.

3.12.3 Separation Between Projectile and Inert Filler.-No separation, visually detectable on radiographic examination, is allowed between the projectile base or sidewall and the inert filler.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-Quality Conformace Inspection shall be conducted on this item.

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4.3 First Article Inspection.-None defined.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain:

a. Ignition element assemblies of one lot interfix number from one manufacturer.

b. Cartridge case components (bodies and bases) of one lot interfix number from one manufacturer.

c. Projectile metal parts of one lot interfix number from one manufacturer.

d. Propellant charges of one lot from one manufacturer.

e. Black powder of one lot from one manufacturer.

f. Tracers of one lot interfix number from one manufacturer.

g. Cartridge case locking rings produced to the same drawing, drawing revisions, specification and specification revisions by one manufacturer.

h. Plug and disc assemblies produced to the same drawing, drawing revisions, specification and specification revision by one manufactuer.

i. Cartridge case mounting rings produced to the same drawing, drawing revision, specification and specification revision by one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Projectile Loading Assembly, Prior to Inserting Fuze Lock Cup				9252343 (If Applicable)
					NEXT HIGHER ASSEMBLY 9266944
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Adhesive missing from fuze lock cup threads of projectiles		100%	3.1	Test
102	Excessive cavitation		0.65% See Note	3.12	4.5.2
103	Incorrect amount of adhesive applied		4.4.3.10	3.1	Balance
Minor	None defined				

NOTES:

MIL-STD-1235, CSP-2 "i" and "f" values as specified by procuring activity.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Projectile Loading Assembly, Prior to Inserting Base Plug and Assembling Windshield Cap		9252343 (If applicable)
			NEXT HIGHER ASSEMBLY 9266944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Adhesive missing from internal threads of fuze lock cup		100%	3.1	Test
102	Adhesive missing from threads of windshield cap		100%	3.1	Test
103	Fuze lock cup fails to withstand min. disassembly torque		0.40	3.3	4.5.3
104	Incorrect amount of adhesive applied		4.4.3.10	3.1	Balance
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9252343 (If applicable)
4.4.2.3	Projectile Loading Assembly				NEXT HIGHER ASSEMBLY 9266944
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Gap between base plug and body excessive		0.40%	3.1	Gage
102	Base plug fails to withstands min. disassembly torque		0.40%	3.3	4.5.3
103	Gap between windshield and windshield cap excessive		0.40%	3.1	Gage
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Assembly, Prior to Assembling Plug and Disc Assembly				9266944
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Tracer missing or inverted		0.40%	3.2	Visual
102	Adhesive missing from plug and disc assembly threads of base plug		100%	3.2	Test
103	Incorrect amount of adhesive applied		4.4.3.10	3.2	Balance
Minor	None defined				

NOTES:

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MIL-C-50902A (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Assembly, Subsequent to Assembly of Plug and Disc Assembly		9266944
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Plug and disc assembly fails to withstand subsequent reapplication of min. assembly torque		0.40%	3.3	4.5.3
102	Projectile weight incorrect, if applicable (See Note)		0.40%	3.1	Scale
Minor					
201	Rotating band damaged		1.0%	3.1	Visual

NOTES:
See Dwg. 9252343

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9266944
4.4.2.6	Assembly of Ignition Element				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Ignition element inserted inversely		100%	3.2	Visual
Major A					
101	Ignition element loose		0.40%	3.2	Manual
102	Ignition element below flush tolerance		0.65%	3.2	Gage
103	Ignition element above flush		0.40%	3.2	Straight-Edge
104	Adhesive missing or incompletely applied		0.40%	3.2	Visual
105	Ignition element circuit continuity over max.		4.4.3.1	3.7	4.5.4
106	Ignition elements deteriorated		4.4.3.2	3.1	MIL-I-50867 (See 6.6)
107	Propellant pin missing, loose, or above flush		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Igniter Assembly, Prior to Loading of Black Powder		9212218
			NEXT HIGHER ASSEMBLY 9266944

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Incorrect weight of igniter charge		0.40%	3.1	Balance
102	Moisture content of black powder over max.		4.4.3.3	3.4	MIL-P-223
Major B					
131	Igniter bag torn, ripped or cut		0.65%	3.1	Visual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9266944
4.4.2.8	Igniter Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Tube cracked or broken		0.65%	3.1	Visual
132	Igniter bag charge missing or inverted		0.65%	3.2	Visual
133	Closing plug missing		0.65%	3.2	Visual
Minor					
201	Adhesive incorrectly applied to tube		1.0%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9266944
4.4.2.9	Assembly, Prior to Assembly of Cartridge Case				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Pre-use inspection of cartridge case components		4.4.3.4	3.10	4.5.5
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9266944
4.4.2.10	Assembly, After Assembly of Cartridge Case				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Thread sealant missing from case locking ring threads		100%	3.2	Visual
102	Case mounting ring not firmly seated		100%	3.2	Manual
103	Incorrect number of case mounting rings assembled		100%	3.2	Visual/Manual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-59902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Assembly, Prior to Assembly of Cartridge Case Base and Prop Charge				9266944
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Case locking ring fails torquing requirements		100%	3.2	Visual (See 6.7)
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 9266944
4.4.2.12	Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical					
1	High temperature*		4.4.3.5.4	3.5.1	4.5.6.3
2.	Safety*		4.4.3.5.7	3.5.2	Visual/Photographic
Special					
3	Smoldering residue*		4.4.3.5.5	3.5.3	4.5.6.6 (See 6.8)
Major A					
101	Excessive residue*		4.4.3.5.5	3.5.3	4.5.6.6 (See 6.8)
102	Muzzle velocity*		4.4.3.5.1	3.5.4	4.5.6.1
103	Tracer functioning*		4.4.3.5.2	3.5.5	4.5.6.2
104	Profile, alignment and max. chamber extraction force		100%	3.2	Gage/Dwg. 9201156 9201157 9201152 9201766
105	Production pull test of cartridge case base		100%	3.2	4.5.7.1

NOTES:

* Proving Ground

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.12	CONTINUATION SHEET				9266944
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
106	Moisture content of cartridge case		4.4.3.7	3.9	Visual/Test (See 6.13)
107	Drying time		4.4.3.6	3.2	4.5.8 (See 6.13)
108	Incomplete, insufficient or incorrect application of adhesive to base (prior to assembly)		100%	3.2	Visual
109	Propelling charge missing (prior to assembly of base)		100%	3.2	Visual
110	Excess adhesive not removed after assembly of base		100%	3.2	Visual
111	Distance from datum diameter on rotating band to base of case		100%	3.2	Gage
112	Improper or incomplete coating of lubrication on rotating band or lubricant not dry to touch		100%	3.2	Visual/Manual
113	Closing torque on clamp		100%	3.2	Visual (See 6.11)
114	Rotating band damaged		100%	3.2	Visual
115	Cartridge case sidewall-base gap incorrect		0.65%	3.2	Gage
116	Assembly time over max.		0.65%	3.2	Visual/Stopwatch

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.12	CONTINUATION SHEET				9266944
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Marking incorrect or unidentifiable including color coding		1.0%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.6	Visual
203	Cartridge case-projectile gap incorrect		1.0%	3.2	Gage
Info Only					
301	Target-impact dispersion*		4.4.3.5.3		4.5.6.2 (See 6.10)
302	Five foot drop test*		4.4.3.5.6		4.5.6.5 (See 6.10)
303	Multiple environmental test*		4.4.3.5.8		4.5.6.4 (See 6.10)
304	Destructive pull test of cartridge case base		4.4.3.8		4.5.7.2

NOTES:

* Proving Ground

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9255465
4.4.2.13	Packing, prior to assembly of upper cushion				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Barrier bag (elastomeric) torn, ripped or missing from cartridge		100%	3.1	Visual
102	Desiccant bag broken, missing or not stitched into a ring		100%	3.1	Visual
103	Cartridge loose within bottom cushion half		100%	3.1	Manual
104	Moisture content of cushion-half		4.4.3.9	3.11	4.5.9
105	Nose support missing		1.0%	3.1	Visual
106	Ignition element ring missing		1.0%	3.1	Visual
107	Forward spacer missing		1.0%	3.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9255465
4.4.2.14	Packing, prior to assembly of container cover				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Incorrect orientation of cartridge within container		100%	3.1	Visual
102	Sealing tape applied improperly		0.65%	3.1	Visual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50902A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Packing				9255465
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Spider assembly arms improperly positioned in bayonet slots		100%	3.1	Visual
102	Contents loose		0.65%	3.1	Manual
103	Container fails to withstand reapplication of min. assembly torque		0.65%	3.3	4.5.3
104	DOD code marking missing incorrect or unidentifiable		0.65%	3.1	Visual
105	Car seal missing, unsealed or improperly positioned		1.0%	3.1	Visual
106	Protective coating damaged or incomplete		1.0%	3.1	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Ignition Element Circuit Continuity.-A sample shall be randomly selected from each shift's production of the ignition element-base subassemblies which are scheduled for use in a complete round lot. The sample size shall be determined from MIL-STD-105, General Inspection Level II. The sample size shall be tested for continuity. Failure of any sample to comply with the applicable requirement shall require 100 percent testing of the ignition element-base subassemblies represented by the failing sample. Any subassembly failing to comply with the requirements shall be rejected and removed from the shift's production.

4.4.3.2 Check Test for Deterioration of Ignition Elements.- If the total elapsed time between original acceptance of an assembled ignition element lot and assembly of that lot into the ammunition components with which it is to be issued exceeds two (2) years, or if the ignition elements have been subjected to adverse conditions at any time since previous tests, or if more than six (6) months have elapsed since a previous check test, the ignition element lots shall be subjected to, and must satisfactorily pass, the check test for deterioration, prior to use.

4.4.3.3 Moisture Content of Black Powder.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. For verification, and just prior to the start of each eight hours loading production, the contractor shall test one sample of black powder selected from that quantity, or sub-lot of powder which is to be used in the eight-hour loading production of ignition element base sub-assemblies. Composite samples shall not be used. If the moisture content of the sample exceeds the requirement and loading has not begun, the sub-lot of powder represented by the sample shall be rejected. If powder from a sub lot with excessive moisture has been used in loading, the remaining unloaded powder and all assemblies containing the non-conforming powder shall be rejected.

4.4.3.4 Pre-use Inspection of Cartridge Case Components.- Just prior to the start of each production lot fifteen cartridge case bodies and bases shall be randomly selected from the cases and bases scheduled for use in the complete round lot. Failure of any sample shall require flaring of the failing sample (s) and reinspection. If reinspection indicates compliance with the applicable requirement the remainder of the cases and bases represented by the samples shall be flared prior to use. If reinspection still indicates rejection, the cases and bases represented by the sample shall not be used in loading.

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4.4.3.5 Proving Ground

4.4.3.5.1 Muzzle Velocity.-

4.4.3.5.1.1 Beginning with the first lot produced and continuing until ten consecutive lots have complied with the applicable requirements specified, thirty (30) samples shall be selected from each lot for this test. The acceptance for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B3, and an AQL of 1.0%. In addition, if the standard deviation of the sample multiplied by the factor 0.76 exceeds the applicable requirement, the lot shall be rejected.

4.4.3.5.1.2 After ten consecutive lots have met the muzzle velocity criteria of 4.4.3.5.1.1 and the last ten consecutive lots have met the residue criteria of 4.4.3.5.5.1, ten (10) samples shall be selected from each lot for this test. The acceptance for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B3, and an AQL of 1.0%. In addition, if the standard deviation of the sample multiplied by the factor 0.65 exceeds the applicable lot requirements, the lot shall be rejected.

4.4.3.5.2 Tracer Functioning

4.4.3.5.2.1 The cartridges fired for muzzle velocity in accordance with 4.4.3.5.1.1 shall be observed for tracer functioning. If four or more samples fail to comply with the applicable requirement, the lot shall be rejected.

4.4.3.5.2.2 The cartridges fired for muzzle velocity in accordance with 4.4.3.5.1.2 shall be observed for tracer functioning. If two or more samples fail to comply with the applicable requirement, the lot shall be rejected.

4.4.3.5.3 Target-impact Dispersion Test.-The cartridges fired for muzzle velocity in 4.4.3.5.1 shall be observed for target impact dispersion.

4.4.3.5.4 High Temperature Test.-A sample of ten (10) cartridges shall be selected from each lot for this test. If any cartridge fails to comply with the applicable requirement, the lot shall be rejected.

4.4.3.5.5 Residue

4.4.3.5.5.1 The cartridges fired for muzzle velocity in accordance with 4.4.3.5.1.1 shall be observed for excessive residue and smoldering residue. The lot shall be rejected for:

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a. Flaming, smoking or smoldering residue in a stationary position on the breech surface which remains in evidence seven seconds or longer after the firing pulse.

b. Flaming, smoking or smoldering residue in a stationary position on the tube surface to the rear of the missile slot which remains in evidence five seconds or longer after the firing pulse.

c. Evidence of two or more rounds exhibiting excessive residue.

The lot shall be referred for:

a. Flaming, smoking or smoldering residue in a stationary position on the breech surface which is extinguished earlier than seven seconds after the firing pulse.

b. Flaming, smoking or smoldering residue in a stationary position on the tube surface to the rear of the missile slot which is extinguished earlier than five seconds after the firing pulse.

c. Any non-stationary flaming, smoking or smoldering residue.

d. Any stationary flaming, smoking or smoldering residue forward of the origin on the missile slot.

4.4.3.5.5.2 The cartridges fired for muzzle velocity in accordance with 4.4.3.5.1.2 shall be observed for excessive residue and smoldering residue. If any lot fails to comply with the applicable residue requirement, or if cartridge cases of a different manufacturing source are introduced, an additional 20 sample cartridges shall be randomly selected from each lot for this test. The additional sample of 20 shall continue until ten consecutive lots have complied with applicable requirements for residue. The lot shall be rejected for:

a. Flaming, smoking or smoldering residue in a stationary position on the breech surface which remains in evidence seven seconds or longer after the firing pulse.

b. Flaming, smoking or smoldering residue in a stationary position on the tube surface to the rear of the missile slot which remains in evidence five seconds or longer after the firing pulse.

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c. Evidence of two or more rounds exhibiting excessive residue.

The lot shall be referred for:

a. Flaming, smoking or smoldering residue in a stationary position on the breech surface which is extinguished earlier than seven seconds after the firing pulse.

b. Flaming, smoking or smoldering residue in a staionary position on the tube surface to the rear of the missile slot which is extinguished earlier than five seconds after the firing pulse.

c. Any non-stationary flaming, smoking or smoldering residue.

d. Any stationary flaming, smoking or smoldering residue forward of the origin on the missile slot.

4.4.3.5.5.3 The cartridges fired in the high temperature test (4.4.3.5.4) shall be observed for excessive and smoldering residue. Any occurrences of excessive or smoldering residue shall be reported.

4.4.3.5.6 Five Foot Drop Test.-From the first production lot, ten cartridges shall be randomly selected and subjected to a 5 foot (unpackaged) drop test.

4.4.3.5.7 Safety.-All cartridges fired in the proving ground tests shall be observed for hangfire and breakup of metal parts. If any cartridge fails to comply with the applicable requirement, the lot shall be rejected.

4.4.3.5.8 Multiple Environmental Test.-From the first production lot, 20 cartridges shall be randomly selected and subjected to a multiple environmental test.

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4.4.3.6 Drying Time.-Each skid of assembled cartridges shall be visually inspected for drying time in the conditioning bay. If any skid document indicates time in the conditioning bay lower than the minimum total requirement, the skid of cartridges represented by the document shall be placed back in the conditioning bay until compliance with the applicable requirements is indicated.

4.4.3.7 Moisture Content of Cartridge Case.-Just prior to removing each skid of cartridges from the conditioning bay after final drying, the relative humidity (RH) and the temperature in the conditioning bay shall be checked and recorded on the skid document. If the RH is above the maximum allowable level or if the temperature does not comply with the applicable requirement, no cartridges will be removed from the conditioning bay. Instead, the RH and/or the temperature in the conditioning bay must be corrected to reach an acceptable level and the moisture test specified in MIL-STD-286B Method 101.4 must be performed on one cartridge selected at random from the skid which was scheduled for removal. This cartridge will represent all cartridges in the conditioning bay at the time the discrepancy was observed. The samples required for the test shall be prepared as follows: Two .025 gm (approximate) samples shall be sectioned from the case body at random locations and two .025 gm (approximate) samples shall be sectioned from the case base. The temperature control shall be set at 100 degrees C with a timer setting of 15 minutes. Cartridges represented by the samples shall remain identified until successful performance during the moisture content test is indicated. Acceptable performance during the moisture test will qualify the skid for removal from the conditioning bay and shall reinstate normal inspection for RH and temperature. Under no circumstances will any cartridges in the conditioning bay be removed until acceptable performance during the moisture test is indicated. If the samples fail to comply with the applicable requirement, all the cartridges in the conditioning bay at the time the discrepent RH and/or temperature readings were observed will remain in the conditioning bay and additional 2 hours + 1/2 hour (in addition to the final curing time specified on the drawing). One hour prior to completion of the additional curing time, samples shall again be taken from the cartridge originally selected and subjected to testing as specified above. Failure of these samples to comply with the applicable requirement shall reject the skid originally scheduled for removal and will require moisture testing as above on one cartridge selected at random from each subsequent skid of cartridges until acceptable performance is demonstrated. Skids indicating non-conformance

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shall also be rejected. Normal inspection may be reinstated coincident with determination of acceptable moisture content. Upon return to normal inspection, all cartridges remaining in the oven and represented by the original moisture sample shall be considered acceptable as long as the deviation from relative humidity (RH) and/or temperature requirements does not exceed the amount of such deviation at the time the acceptable moisture sample was taken.

4.4.3.8 Destructive Pull Test.-Inert pull test cartridges (three total) shall be loaded at random intervals with production cartridges in a lot. The three inert pull test cartridges shall be assembled with only the cartridge case body, mounting ring, locking ring (no thread sealant) and cartridge case base, using adhesive, and shall be conspicuously identified. After final drying, the three pull test cartridges shall be subjected to a pull test to destruction of the cartridge case base. Results as to force required prior to separation shall be forwarded to Picatinny Arsenal, ATTN: SARPA-QA-A-A.

4.4.3.9 Moisture Content of Cushion-half.-The sampling plan and procedures for this test shall be in accordance with MIL-STD-1235, using CSPI inspection level III with an AQL of 0.40 percent. Any sample cushion-half which fails to meet the requirement, shall be considered a reject. Rejected cushion-halves may be dried in a circulating air oven at 100°F to 150°F and resubmitted for acceptance. Production interval shall be the cushion-half vendor's lot. This test replaces the receiving activity moisture testing required by Specification MIL-P-60312.

4.4.3.10 Adhesive Applicators.-The automated thread sealant applicators will be checked on an hourly basis to make certain the amount of adhesive specified on the applicable drawing is being dispensed. Failure to comply with the drawing requirement shall be cause for rejection of the previous hour's production.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 of this specification.

4.5 Test Method and Procedures

4.5.1 Materials, Components and Processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of

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inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspections.-The visual, manual, and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein.

4.5.2 Determination of Cavities.-The sample projectiles shall be subjected to x-ray examination. In all exposures, the x-ray beam shall form a 90° angle with the longitudinal axis of the projectile. Examination of the x-ray nagative shall be made tb determine compliance with applicable requirements. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. In order to obtain good definition of the area at the base of the projectile, two separate speeds of film or exposure times may be required. In addition, the sensitivity of x-ray techniques shall be 2.0%, and the photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.5.2.1 and 4.5.2.2. In the event an original x-ray picture leaves doubt as to acceptability, the projectile shall be re-x-rayed, at an angle 90° from that of the original exposure.

4.5.2.1 Check Sensitive of X-Ray Technique.-The sensitivity of x-ray technique shall be determined by means of penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the projectile away from the film (source side). Observation shall be made of the image of the penetrameter on the x-ray film. The sensitivity of x-ray technique shall be considered satisfactory when the image of the penetrameter, including the 2T hole, is sharply defined on the x-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thickness and shall be equal to 2.0% of the maximum equivalent thickness. The penetrameter shall have a hole, the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter, see MIL-STD-453. The actual thickness may be determined from a radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated in Figure 2 of this specification.

4.5.2.2 Check of Photographic Density of Film.-The photographic density of the film shall be determined on the film in a region of the sound charge plus metal corresponding to that portion of the material which is of the greatest combined thickness to be

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examined on the negative. To judge densities, it is recommended that standards of radiographic densities be used. Comparison of film with standards may be made visually; however, the use of a densitometer is recommended.

4.5.3 Torque Testing.-The torque shall be applied in a tightening direction for assembly torques, and in a loosening direction for disassembly torques, while holding the parts immediately adjacent to the joints being tested so as to avoid possible loosening of other joints. The joint under test shall be marked to provide a means for detection of any movement between the adjacent parts. Any discernible movement between adjacent parts shall be regarded as torque failure.

4.5.4 Ignition Element Circuit Continuity.-The ignition element assembly, after insertion into the cartridge case base, shall be tested for resistance by the following method: A probe shall be inserted through the test holes of the base and ignition element to the inner sleeve of the ignition element and the circuit completed by connecting a lead from the probe to the outer cup of the ignition element. An ohmmeter shall be attached and the resistance read to the nearest 0.1 ohm. No more than 10 milli-amperes current shall be induced in the circuit during this test.

4.5.5 Pre-use Inspection of Cartridge Case Components.-The cartridge case base shall be assembled to the cartridge case body (without glue) and the base clamp assembled and torqued to the required value. A .010 inch thick by .375 inch wide feeler gage shall be used to check the gap between the outside diameter of the cartridge case body and the inside diameter of the cartridge case base. The gap shall be checked at four locations, 90° apart. The gage shall not enter to a depth exceeding .0625 inch at any point.

4.5.6 Proving Ground Tests.-These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedures. The sample cartridges shall be fired in a gun for which the cartridge is standard without a closed breech bore scavenger utilizing a 110 volt DC power source (See 6.12) and observation shall be made for con-formance to all the requirements.

4.5.6.1 Velocity-target Impact Dispersion.-The ammunition for this test shall be fired against reference rounds after being conditioned at a temperature of 70°F±2°F for a period of 16 hours prior to firing. The cartridge shall be fired at a cloth target placed at a distance of 2000 meters from the muzzle

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of the gun. The hit pattern results in vertical and horizontal probable error in mils shall be recorded on the firing record for informational purposes. Muzzle velocity shall be determined and recorded for each round.

4.5.6.2 Tracer Functioning.-Cartridges fired during the velocity test shall be observed for tracer functioning to a distance of 1000 meters. The distance over which the tracer burns shall be recorded to the nearest estimated 250 meters (i.e. failure at 250 meters, 500 meters, 750 meters).

4.5.6.3 High Temperature Test.-Cartridge for this test shall be conditioned in such a manner that the cartridge case with propellant charge, elastomeric bag assembled, is exposed to the conditioning environment. The conditioning environment shall be 125°F (± 2°F) and the rounds shall be exposed for sufficient time to allow the propellant temperature to stabilize at 125°F (± 2°F) (16 hours min. to 24 hours max.). Pressure gage for this test shall be conditioned and maintained at 70°F (± 2°F) and shall be placed in the cartridges not more than five minutes before firing. After conditoning, the cartridges shall be fired at +125°F. Chamber pressures shall be recorded and muzzle velocities shall be measured for information. Smear cameras shall be positioned to record the initial portion of projectile flight, as a minimum. Observation shall be made for separation of parts likely to result in a safety hazard.

4.5.6.4 Multiple Environmental Test.-The 20 samples cartridges shall be divided into two groups of 10 each, designated as Group A and Group B. Each group shall be conditioned and tested as indicated below.

4.5.6.4.1 Group A.-The conditioning and test procedures for Group A cartridges is as follows:

a. Condition the cartridges as shipped for a minimum of 48 hours at 145°F and 95% relative humidity (RH).

b. Condition the packed cartridges at -65°F for a minimum of 48 hours.

c. Condition the packed cartridges (a second time) at 145°F and 95% RH.

d. While maintaining the cartridge at 145°F, conduct a transportation vibration test with the packaged cartridges in accordance with MTP-4-2-804.

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e. Remove the cartridge from its packing.

(1) Record any unusual difficulty in removing cartridges from container.

(2) Record any signs of deterioration or corrosion of packing and packaging material, cartridge case and projectile.

f. The cartridges shall be fired for velocity-target impact dispersion as in 4.3.2.1 with the exception that firing temperature shall be 125°F. Chamber pressure shall also be recorded.

4.5.6.4.2 Group B.-The conditioning and testing of Group B cartridges shall be as indicated in 4.5.6.4.1, with the exception that sequence (c), second high temperature conditioning, shall be omitted; sequence (d), transportation vibration, shall be conducted at a round temperature of -65°F and sequence (f), ballistic firing, shall be conducted at a round temperature of -40°F.

4.5.6.5 Five Foot Drop Test.-Five cartridges shall be conditioned at -65°F and five cartridges shall be conditioned at 145°F for eight hours. Following the temperature conditioning, the test shall be conducted in accordance with MIL-STD-331, Test Method 111. Observation shall be made for spilled propellant.

4.5.6.6 Residue.-Camera coverage of the breech chamber cavity, tube chamber and bore is required immediately after firing and upon opening of the automatically operated breech for all cartridges fired, except those fired in accordance with paragraph 4.4.3.5.8. Camera coverage will be used to determine compliance with the smoldering residue requirement. In addition, excessive residue observed during and firings shall be collected, identified and photographed.

4.5.7 Pull Test of Cartridge Case Base

4.5.7.1 Production.-This test shall be performed after the bond at the cartridge case body-base joint has had sufficient time to set. The cartridge shall be held in a suitable test fixture clamping on the projectile body. The test apparatus shall apply a longitudinal load of 150 pounds distributed over 360° around the lip of the cartridge case base. Observation shall be made for separation of the cartridge case base from the cartridge. The method of test should not damage the ammunition components and paint or markings damaged during the test shall be restored. Cartridges which meet the requirement may be returned to the lot provided they have not been damaged by the method of test.

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4.5.7.2 Destruction.-This test shall be performed after the bond at the cartridge case body/base joint has had sufficient time to set. The cartridge shall be held in a suitable test fixture clamping on the projectile body. The test apparatus shall apply a longitudinal load distributed over 360° around the lip of the cartridge case base. Observation shall be for the force required for separation of the cartridge case base from the cartridge.

4.5.8 Drying Time.-For each skid of cartridges, the document representing the skid shall be annotated with the time and date that the skid of cartridges entered the conditioning bay and the time and date that the skid of cartridges was removed from the conditioning bay.

4.5.9 Moisture Content of Cushion-half.-Moisture testing of sample cushion-halves shall be accomplished using the following equipment and procedures or equivalent:

G8 Master Measuring Unit
Sensitivity Module No. 700
Special Electrode No. 3064

products of Moisture Register Co., Alhambra, Calif. The test procedures shall consist of passing the electrode across the entire surface of the sample cushion-half and observing the meter reading. A meter reading in excess of nine scale divisions at any point on the sample surface indicates moisture content in excess of the requirement. Calibration procedures and intervals for the test equipment will be submitted for approval to Picatinny Arsenal, SARPA-QA-A-A.

5. Preparation for Delivery

5.1 Packaging.-Packaging shall be as specified on Dwg. 9255465.

5.2 Packing and Marking.-Packing and marking shall be as specified on Dwg. 9255465.

6. Notes

6.1 Intended Use.-The cartridge covered by this specification is intended for use with the M60A2 Tank (M162 Gun) and with the M551 Armored Reconnaissance/Airborne Assault Vehicle (M81 Gun).

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6.2 Ordering Data.-See MIL-A-48078 (PA).

6.3 Submission of Designs for Approval.-See 6.2.3 of MIL-A-48078 (PA). Submit equipment designs, as required, to:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-T-T
Dover, New Jersey 07801

6.4 Inspection Lot Size.-It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of test or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation critieria and sampling provisions of this specification are maintained.

6.5 Visual Examination.-When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.6 Cost of Check Test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.7 Torquing of Case Locking Ring.-Due to the compressibility of the cartridge case material during the assembly of the case body to the projectile, the case locking ring will lose its torque almost immediately. Inspection for compliance with the applicable torquing requirement is therefore limited to witnessing the torquing operation.

6.8 Residue

6.8.1 Excessive Residue.-Excessive residue shall be defined as residue large enough to prevent chambering or otherwise hinder functioning of a subsequent round. In addition, any single piece of residue larger than 1 inch by 2 inches should be considered excessive. Any collected residue and photographs shall be forwarded to Picatinny Arsenal, SMUPA-QA-A-A for evaluation.

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6.8.2 Smoldering Residue.-All movies of smoldering residue shall be forwarded to Picatinny Arsenal, SARPA-QA-A-A for evaluation.

6.9 Submission of Test Data and Ammunition Data Cards.-In addition to the normal distribution of records, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commanding Officer, Picatinny Arsenal, ATTN: SARPA-QA-A-A, SARPA-AD-E-A-1, SARPA-KI, Dover, New Jersey, 07801.

6.10 Informational Test Data.-Test results for the "information only" type ballistic tests need not be submitted at the same time as the acceptance test report.

6.11 Torquing of Base Clamp.-Due to the requirement prohibiting reapplication of assembly torque, inspection for compliance with the applicable torquing requirement is limited to witnessing the torquing operation.

6.12 Misfires.-If a misfire not related to the weapon firing system is experienced during any phase of proving ground testing with a 110 volt DC power source, the misfired round(s) shall be held for investigation, and the lot shall be referred for disposition.

6.13 Skid Documents.-Documentation on relative humidity and temperature in the conditioning bay as well as drying time shall be retained for each skid until the lot which the documents represent has been finally accepted.

6.14 Gluer Qualification .-Since the gluing operation associated with the assembly of the cartridge case base to the cartridge case body is extremely important to overall cartridge case strength, it is recommended that personnel associated with this operation be carefully trained. A gluer qualification test is suggested. Properly assembled cartridge case body to base joints should be capable of withstanding 1000 pounds of force during pull testing.

6.15 Proving Ground Test Summary: See Next Page.

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ARMY-PA

Preparing Activity:
ARMY- PA

Project Number: FSC: 1320-A- 249

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PROVING GROUND TEST SUMMARY

TEST	SAMPLE SIZE		REQUIREMENTS		SAMPLING PARAGRAPH	TEST PROCEDURES PARAGRAPH NUMBER
	INITIAL (NOTE 1)	SUBSEQUENT				
Muzzle Velocity	30	10	a.	Muzzle Velocity (3.5.4)	4.4.3.5.1	4.5.6.1
			b.	Tracer Functioning (3.5.5)	4.4.3.5.2	4.5.6.2
			c.	Safety (3.5.2)	4.4.3.5.7	-------
		(note 2)	d.	Residue (3.5.3)	4.4.3.5.5	4.5.6.6
			e.	Target Impact Dispersion (Info)	4.4.3.5.3	4.5.6.1
HIGH TEMPERATURE	10	10	a.	Pressure (3.5.1)	4.4.3.5.4	4.5.6.3
			b.	Safety (3.5.2)	4.4.3.5.7	-------
			c.	Muzzle Velocity (Info)	4.4.3.5.4	4.5.6.3
			d.	Residue (Info)	4.4.3.5.5	4.5.6.6

NOTE 1: Until ten consecutive lots have been accepted for muzzle velocity and the last ten consecutive lots have been accepted for residue.

2: An additional 20 items are required for residue testing if a lot on subsequent sampling is rejected for residue.

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PROVING GROUND TEST SUMMARY (Continued)

Multiple Environmental	20 (NOTE 3)	a.	Muzzle Velocity (Info)	4.4.3.5.8	4.5.6.4
		b.	Pressure (Info)	4.4.3.5.8	4.5.6.4
		c.	Target Impact Dispersion (Info)	4.4.3.5.8	4.5.6.4
		d.	Safety (3.5.2)	4.4.3.5.7	-------
Five Foot Drop	10 (NOTE 3)		Safety (3.5.2)	4.4.3.5.6	4.5.6.5

NOTE 3 : First lot only

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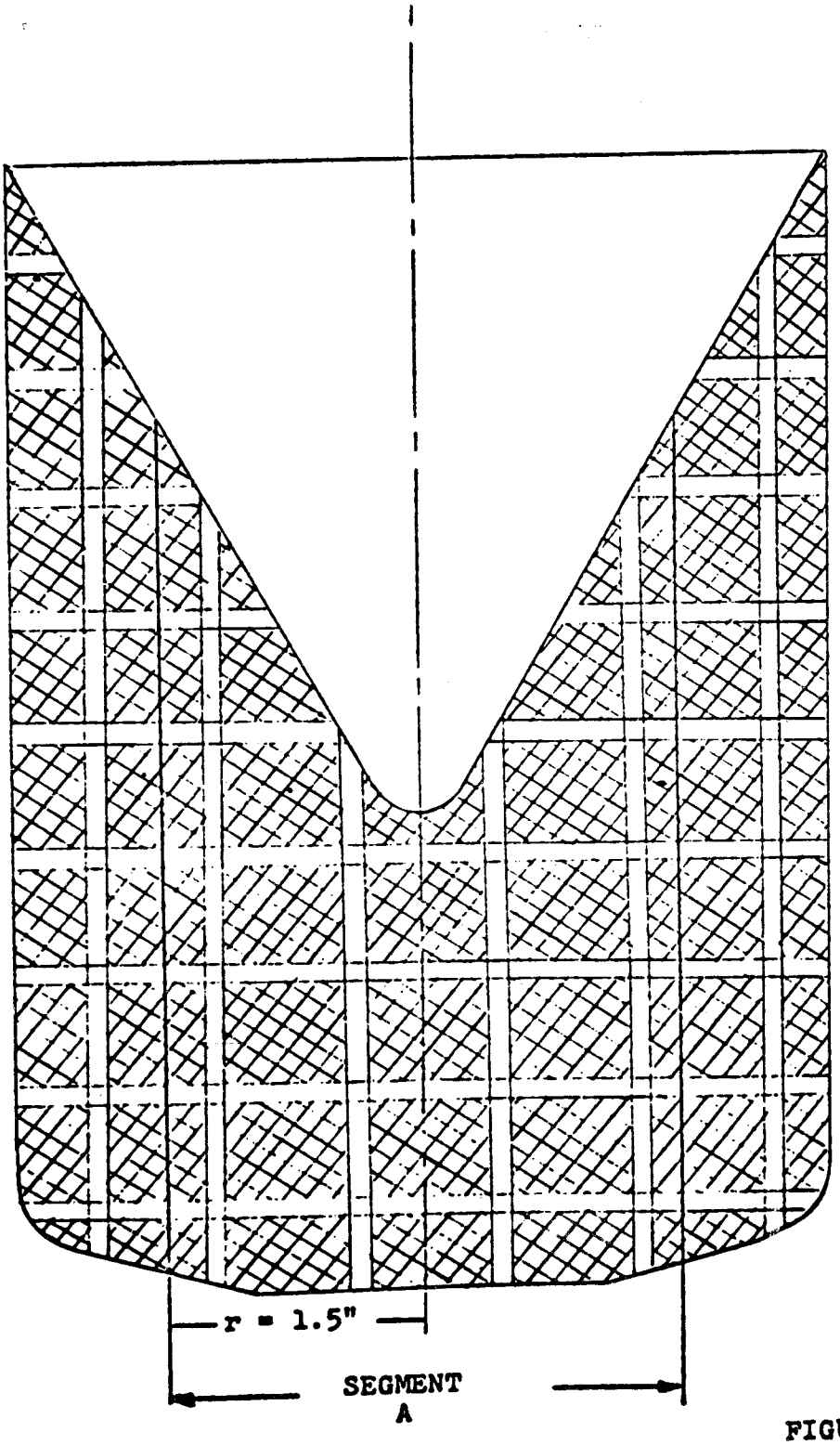


FIGURE 1

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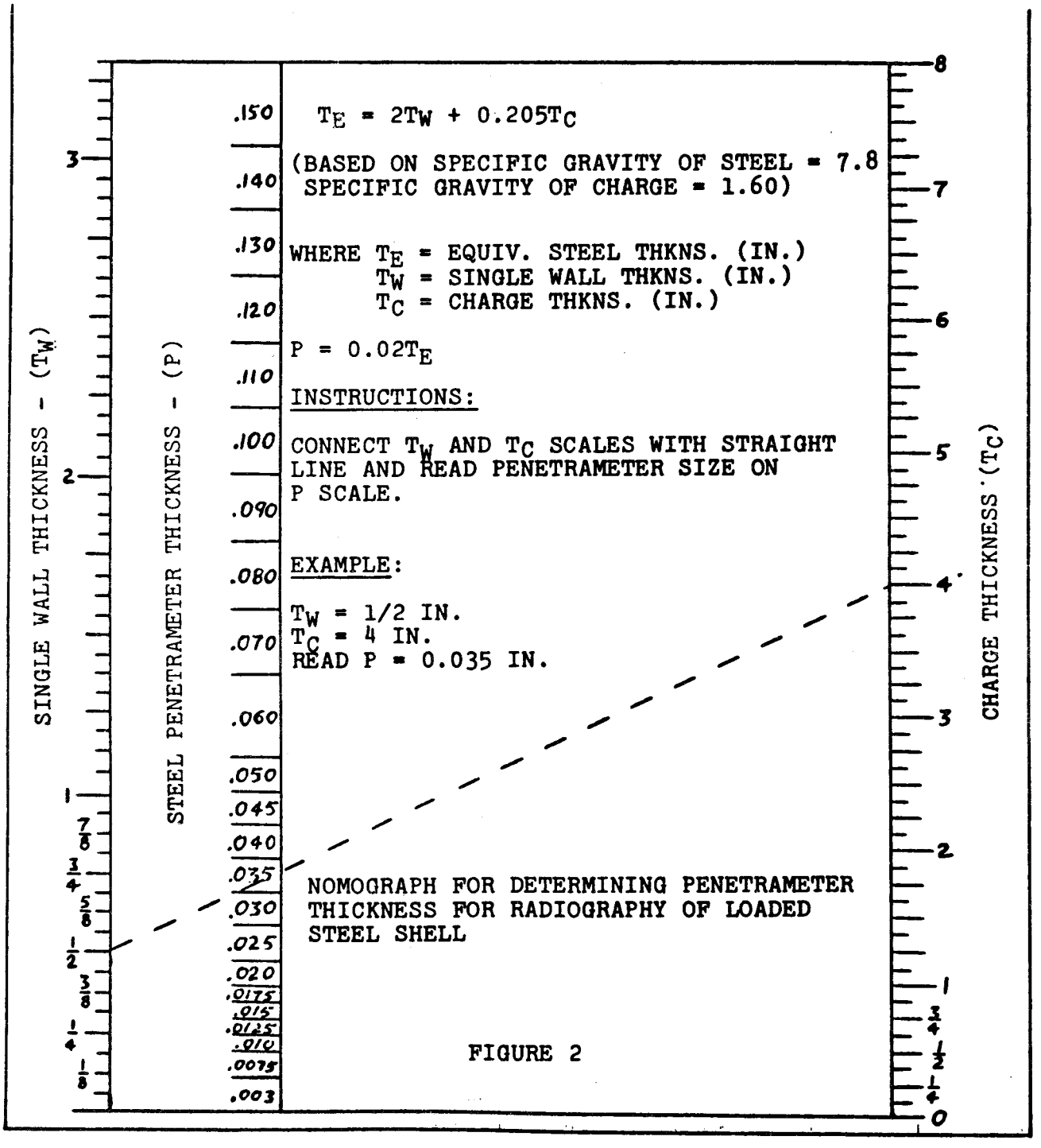


FIGURE 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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